

Data Wrangling and Data Analysis

Exploratory Data Analysis

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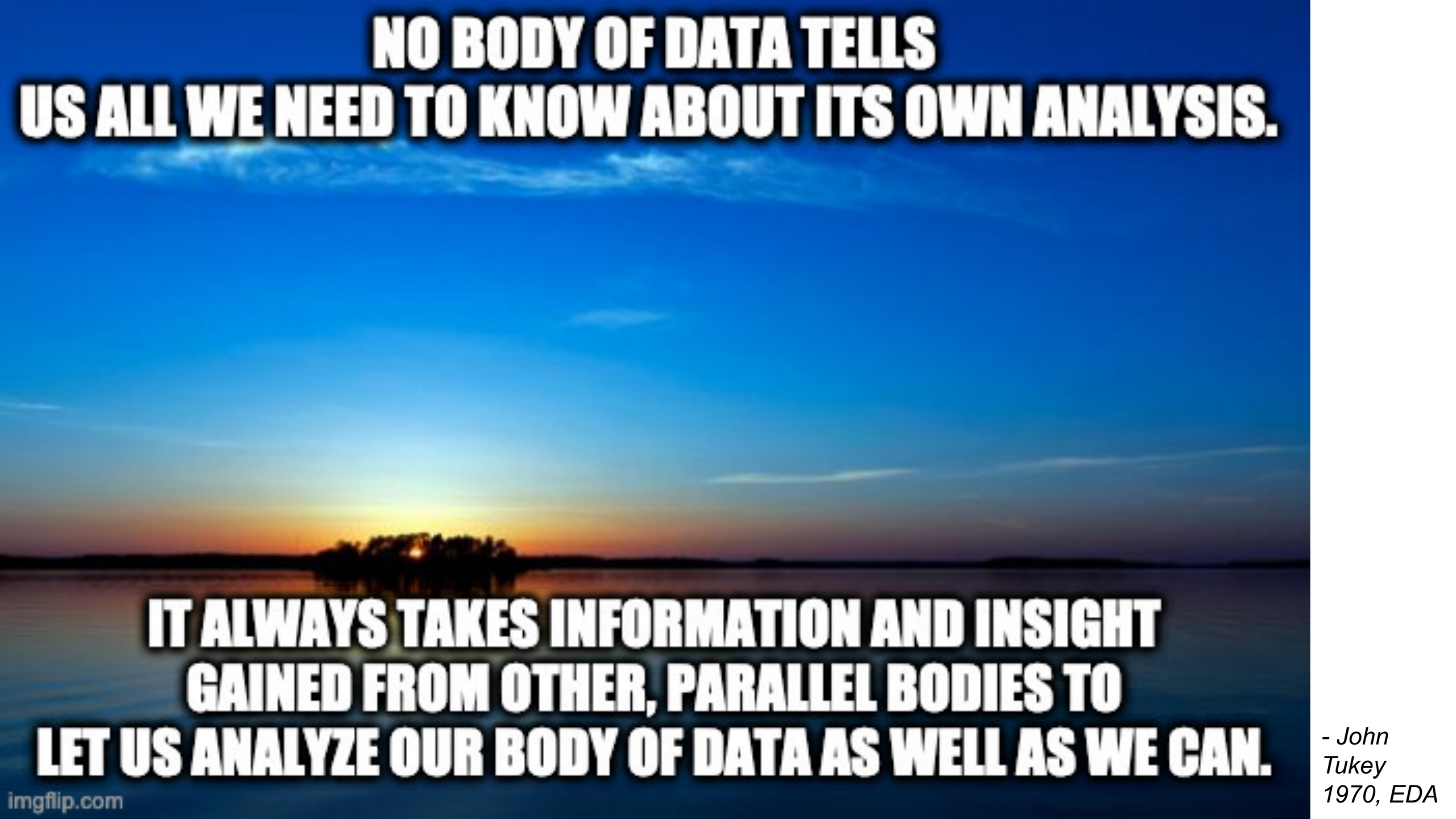
Utrecht University





John Tukey (1915 – 2000)

- Data Scientist patient zero
- Inventor of:
 - The boxplot
 - The term “exploratory data analysis”
 - The Fast Fourier Transform
 - “Tukey’s test”
 - The word “bit”
 - So, so much more (Wikipedia)



**NO BODY OF DATA TELLS
US ALL WE NEED TO KNOW ABOUT ITS OWN ANALYSIS.**

**IT ALWAYS TAKES INFORMATION AND INSIGHT
GAINED FROM OTHER, PARALLEL BODIES TO
LET US ANALYZE OUR BODY OF DATA AS WELL AS WE CAN.**

This week

- Data visualization principles
 - The grammar of graphics
 - **Exploratory data analysis (EDA)**
-
- Goal of the week: know how to create and improve data visualizations, and know how to use them for exploration

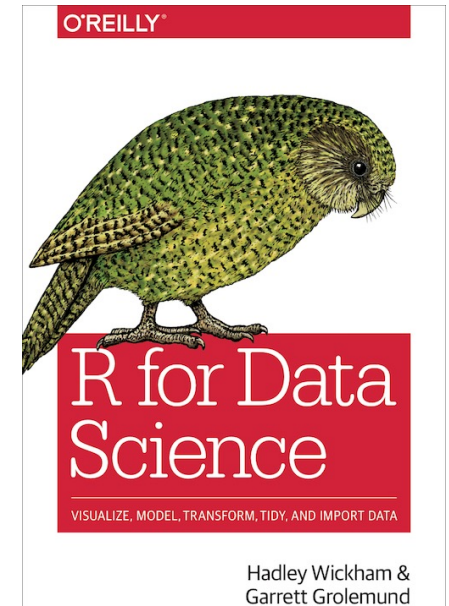


Reading materials for this week

- **R for Data Science (R4DS)**
(*Wickham & Grolemund 2017(ish)*)
- <https://r4ds.had.co.nz>
- Chapters 3,5, 7

Optional:

- **Exploratory Data Analysis with R** (*Peng 2020*)
- <https://bookdown.org/rdpeng/exdata/>



Exploratory Data
Analysis with R



Roger D. Peng



Assignments this week

- Monday: Tidy data visualization with dplyr and ggplot
- Tuesday: Exploratory data analysis
- Thursday: either (a) resit for the test, or (b) assignment on advanced data visualization



Exploratory vs. confirmatory

Exploratory vs confirmatory

- Exploratory analysis
 - Generate new insights
 - Create new hypotheses
 - Analysis may depend on data
- Confirmatory analysis
 - Test theory
 - A priori hypothesis
 - Analysis predefined
 - Example: registered report



Exploratory vs confirmatory

You work at Google and your colleague has implemented a new search algorithm for German searches. Your task is to check whether this algorithm performs better than the existing one.

- **Confirmatory.**
- Theory testing, hypothesis: new works better than old
- Analysis can be defined in advance: which outcome variables, how to sample from the population, which method?
- Full analysis script could be written before the data even exists



Exploratory vs confirmatory

You have obtained access to your company's customer relations database. Your task is to find data-driven ways in which your company can improve customer retainment.

- **Exploratory.**
- Generate new insights - at which touchpoint do customers drop out?
- Create new hypotheses - there may be two types of customers who drop out
- Analysis cannot be defined in advance, task is to explore associations between features



Exploratory vs confirmatory

- Both are necessary and complementary
- Typical scientific questions: exploratory -> confirmatory
- Then build on the tested theories to generate new insights: confirmatory -> exploratory?
- This lecture: **exploratory analysis**





DOUGLAS ADAMS

The Hitchhiker's Guide to the Galaxy



a novel

Some good advice on data exploration

Peng's EDA checklist

1. Formulate your question
2. Read in your data
3. Check the packaging, (run `str()`)
4. Look at the top and the bottom of your data
5. Check your “n”s
6. Validate with at least one external data source
7. Try the easy solution first
8. Challenge your solution
9. Follow up

Exploratory Data
Analysis with R



Roger D. Peng



Tukey's approach to EDA

- Look at center and spread
- Find comparisons
- “Straightening and flattening”:
 - Use logarithms and other transforms
 - Use models and residuals



hinges.

If we have 9 values in all, the 5th from either end will be the median, since $\frac{1}{2}(1 + 9) = 5$. Since $\frac{1}{2}(1 + 5) = 3$, the third from either end will be a hinge. If we have 13 values, the 7th will be the median--and the 4th from each end a hinge. In folded form, a particular set of 13 values appears as follows:

-3.2		1.5		9.8
-1.7		1.2	1.8	6.4
-0.4	0.3		2.4	4.3
	0.1		3.0	

The five summary numbers are, in order, -3.2, 0.1, 1.5, 3.0, and 9.8, one at each folding point.

We usually symbolize the 5 numbers (extremes, hinges, median) that make up a

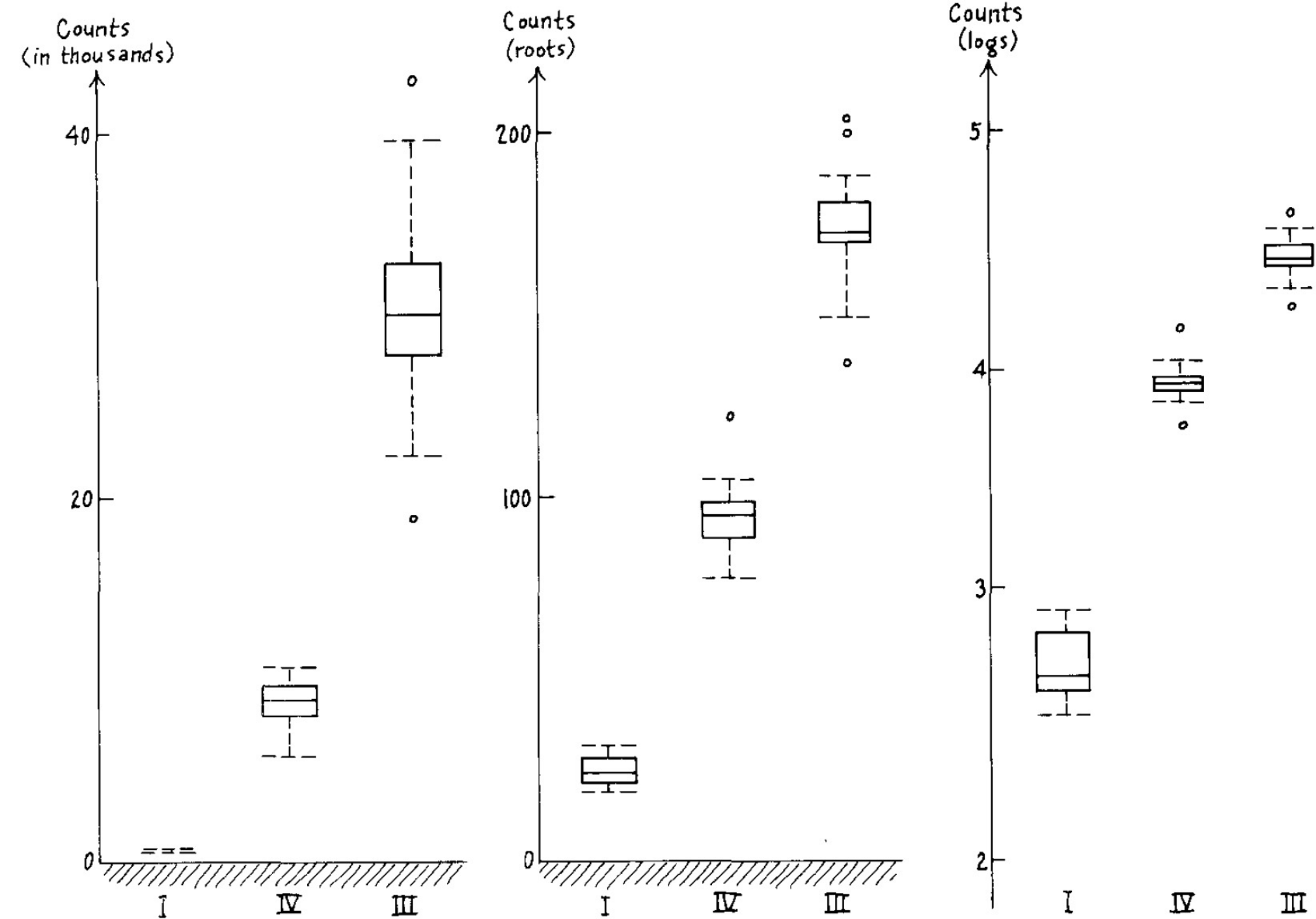
5-number summary

by a simple summary scheme like this:

#13

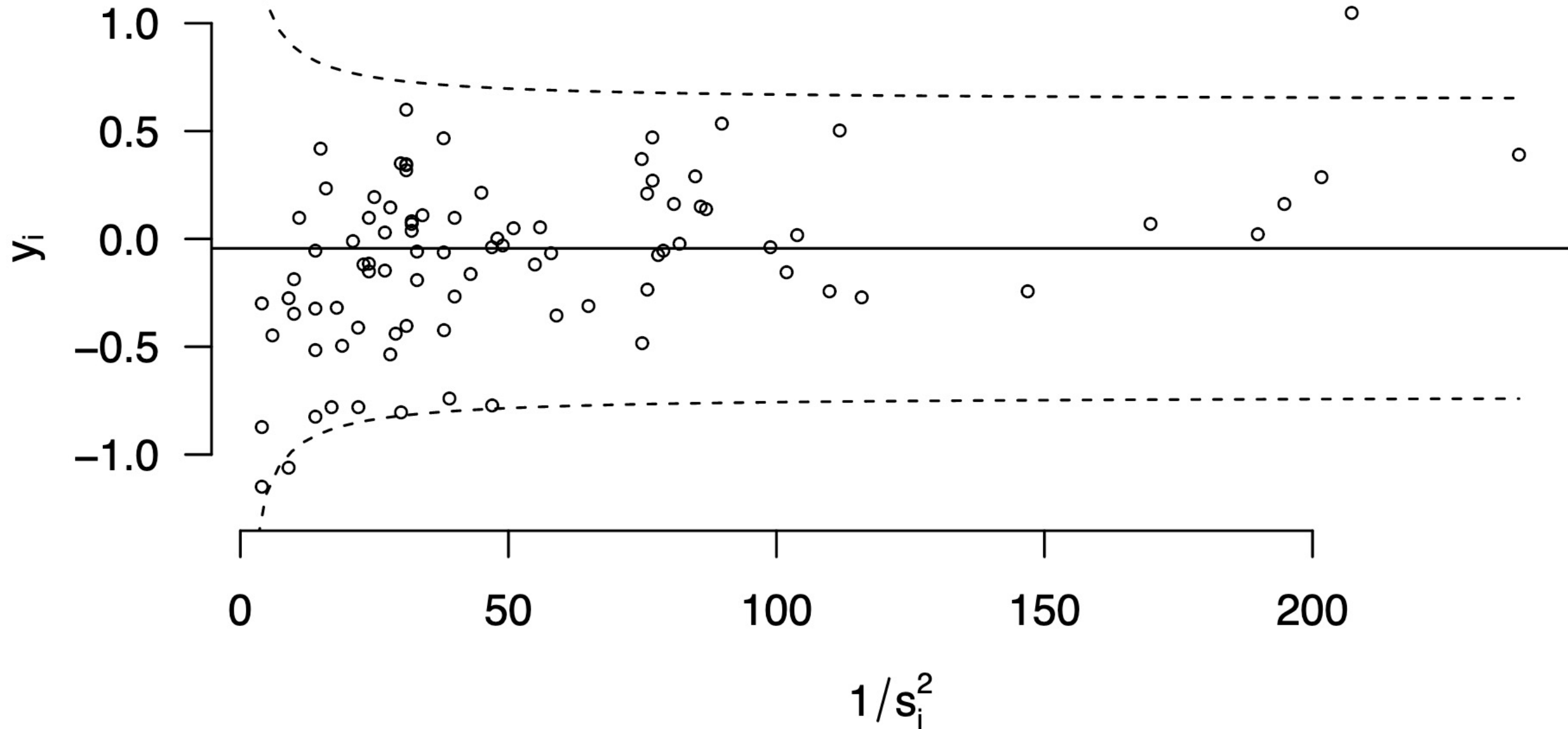
M7	1.5	
H4	0.1	3.0
1	-3.2	9.8

(Tukey 1970)



(Tukey 1970)

Adverse events in Dutch hospitals *(Lenz & Oberski)*



Example exploration of current corona cases in the Netherlands

<https://github.com/J535D165/CoronaWatchNL>

Dataset: COVID-19 case counts in The Netherlands

CoronaWatchNL collects numbers on COVID-19 disease count cases in **The Netherlands**. The numbers are collected from various sources on a daily base, like [RIVM \(National Institute for Public Health and the Environment\)](#), [LCPS \(Landelijk Coördinatiecentrum Patiënten Spreiding\)](#), [NICE \(Nationale Intensive Care Evaluatie\)](#), and the [National Corona Dashboard](#). This project standardizes, and publishes data and makes it **Findable, Accessible, Interoperable, and Reusable (FAIR)**. We aim to collect a complete time series and prepare a dataset for reproducible analysis and academic use.



Read data, check packaging

```
url_icu <-  
"https://raw.githubusercontent.com/J535D165/CoronaWatchNL/master/data-  
ic/data-nice/NICE_IC_long_latest.csv"
```

```
icu <- read_csv(url_icu)
```

```
> dim(icu)  
[1] 1935    3
```



Look at top and bottom

```
> icu %>% tail
# A tibble: 6 x 3
  Datum      Type                               Aantal
  <date>     <chr>                               <dbl>
1 2020-09-28 Toename ontslag (ziekenhuis)           0
2 2020-09-28 Totaal opnamen (IC)                 144
3 2020-09-28 Toename opnamen (IC)              7
4 2020-09-28 Totaal ontslag (IC)              56
5 2020-09-28 Cumulatief opnamen (IC)         3261
6 2020-09-28 Toename ontslag (overleden)       0
```



Look at top and bottom

```
> icu %>% head
# A tibble: 6 x 3
  Datum      Type      Aantal
  <date>    <chr>    <dbl>
1 2020-02-27 Totaal ingezette IC's      5
2 2020-02-27 Cumulatief opnamen (IC)  7
3 2020-02-27 Totaal ontslag (IC)      0
4 2020-02-27 Cumulatief ontslag (ziekenhuis)  0
5 2020-02-27 Cumulatief ontslag (overleden)  0
6 2020-02-27 Toename ontslag (overleden)  0
```



Validate with external source



CoronaTracker

[Home](#)

[Travel Alert](#)

[What is COVID-19](#)

Netherlands Overview

Share: [f](#) [t](#)

114,540

Confirmed

+2,914 new cases

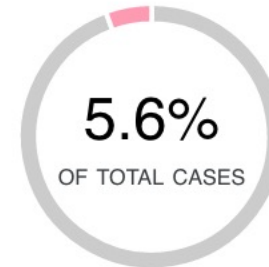
3

Recovered

6,380

Deaths

+6 new deaths



Fatality

Critical Cases treated in ICU

144

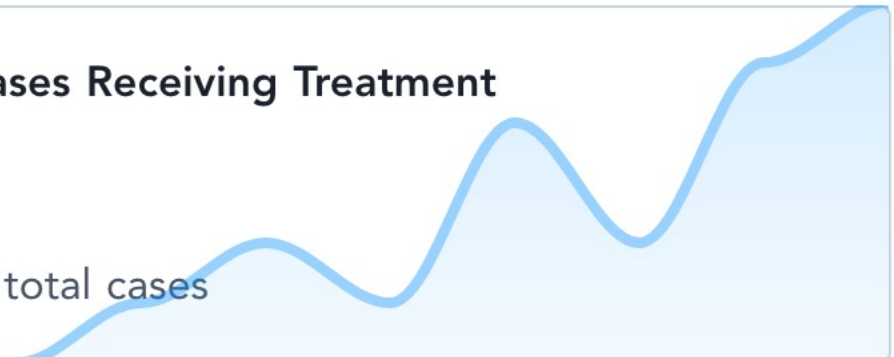
0.1% of total cases



Daily Cases Receiving Treatment

0

0.0% of total cases



Check n's

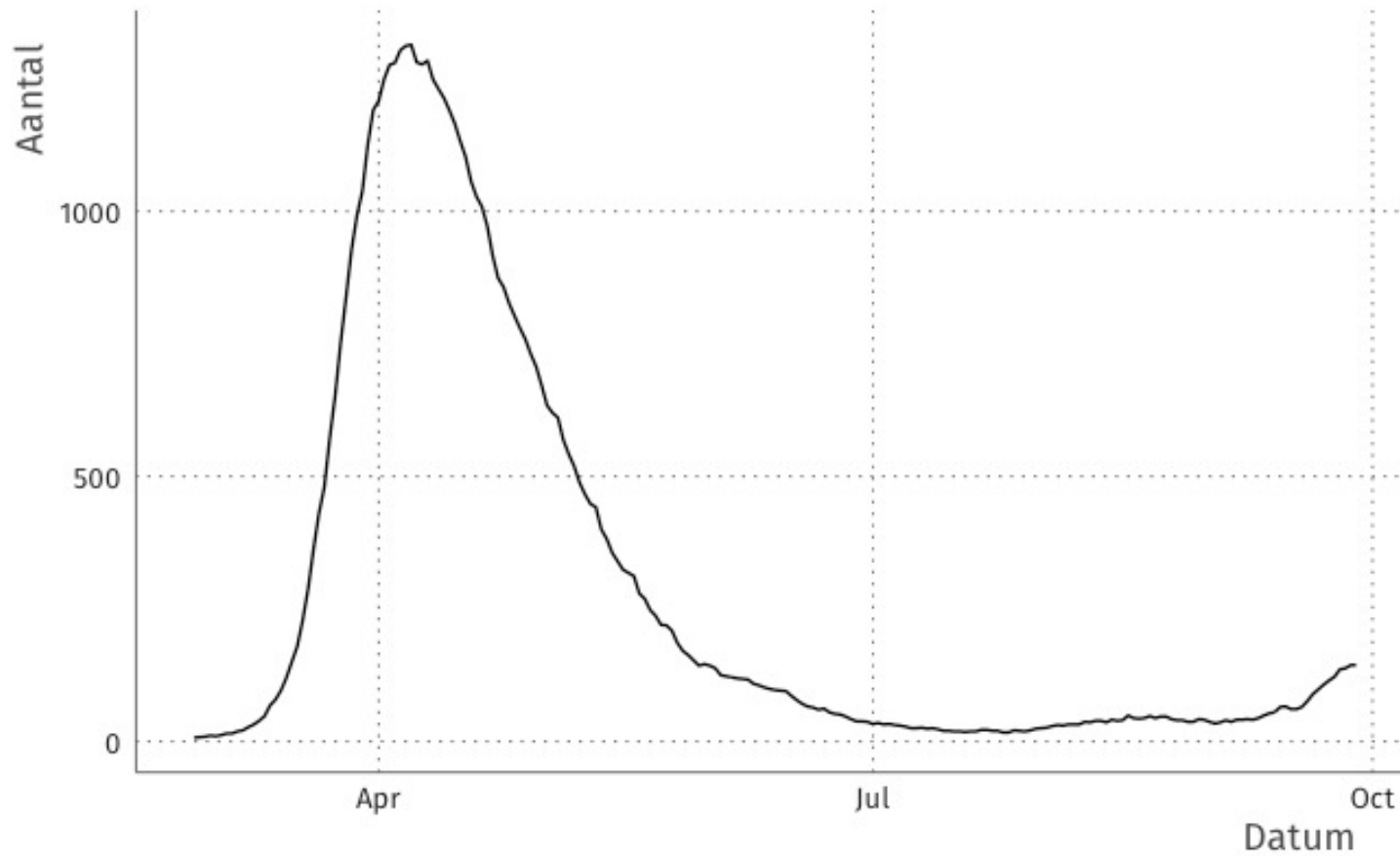
```
> icu %>% group_by(Type) %>% summarize(n())
```

```
# A tibble: 9 x 2
```

Type	`n()`
<chr>	<int>
1 Cumulatief ontslag (overleden)	215
2 Cumulatief ontslag (ziekenhuis)	215
3 Cumulatief opnamen (IC)	215
4 Toename ontslag (overleden)	215
5 Toename ontslag (ziekenhuis)	215
6 Toename opnamen (IC)	215
7 Totaal ingezette IC's	215
8 Totaal ontslag (IC)	215
9 Totaal opnamen (IC)	215



```
icu %>%  
  filter(Type == "Totaal opnamen (IC)") %>%  
  ggplot(aes(Datum, Aantal)) + geom_line() +  
  theme_fira()
```



Understand what you're looking at

Bezetting IC units

		IC-units in Nederland		
		gem.	min.	max.
	bedden	14	1	35
	intensivisten (in fte)	7	1	18
	verpleegkundigen (in fte)	50	10	152

(84 ICUs)

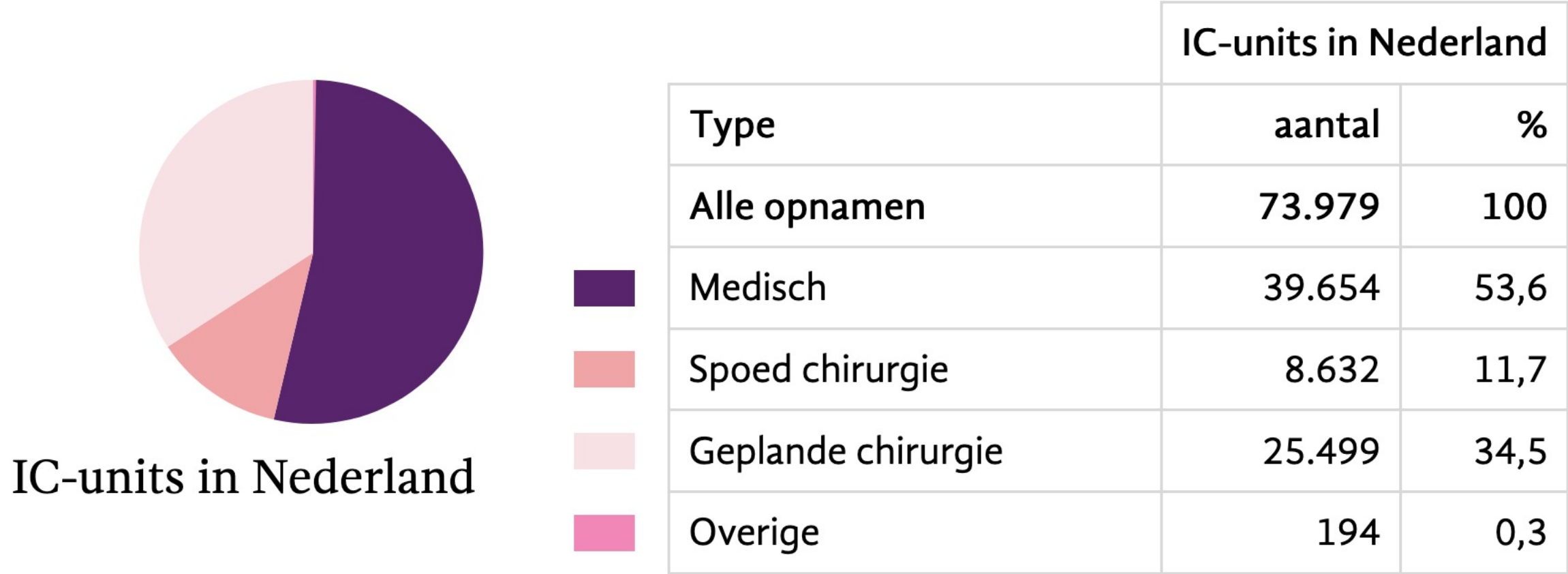
Back-of-the-envelope:

$14 \times 84 \approx 1200$ beds

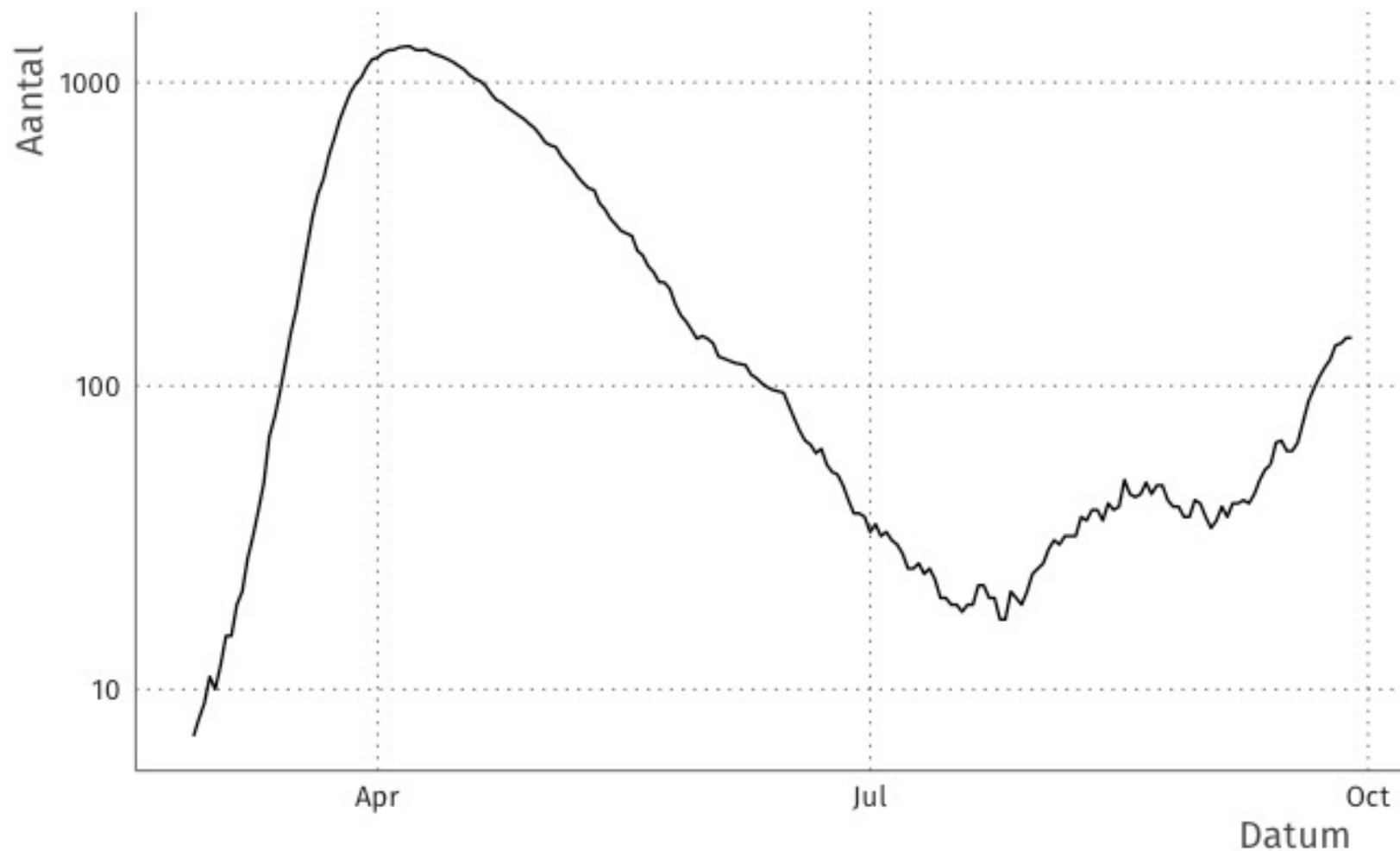
Geometric average:
 $\sqrt{1 \times 35} \approx 6$ beds

Understand what you're looking at

Aantal opnamen per opnametype




```
icu %>%  
  filter(Type == "Totaal opnamen (IC)") %>%  
  ggplot(aes(Datum, Aantal)) + geom_line() +  
  theme_fira() + scale_y_log10()
```



```
url_sewage <-  
"https://raw.githubusercontent.com/J535D165/CoronaWatchNL/master/data-  
dashboard/data-sewage/RIVM_NL_sewage_counts.csv"  
  
sewage <- read_csv(url_sewage)  
  
joined <- bind_rows(sewage, icu)
```



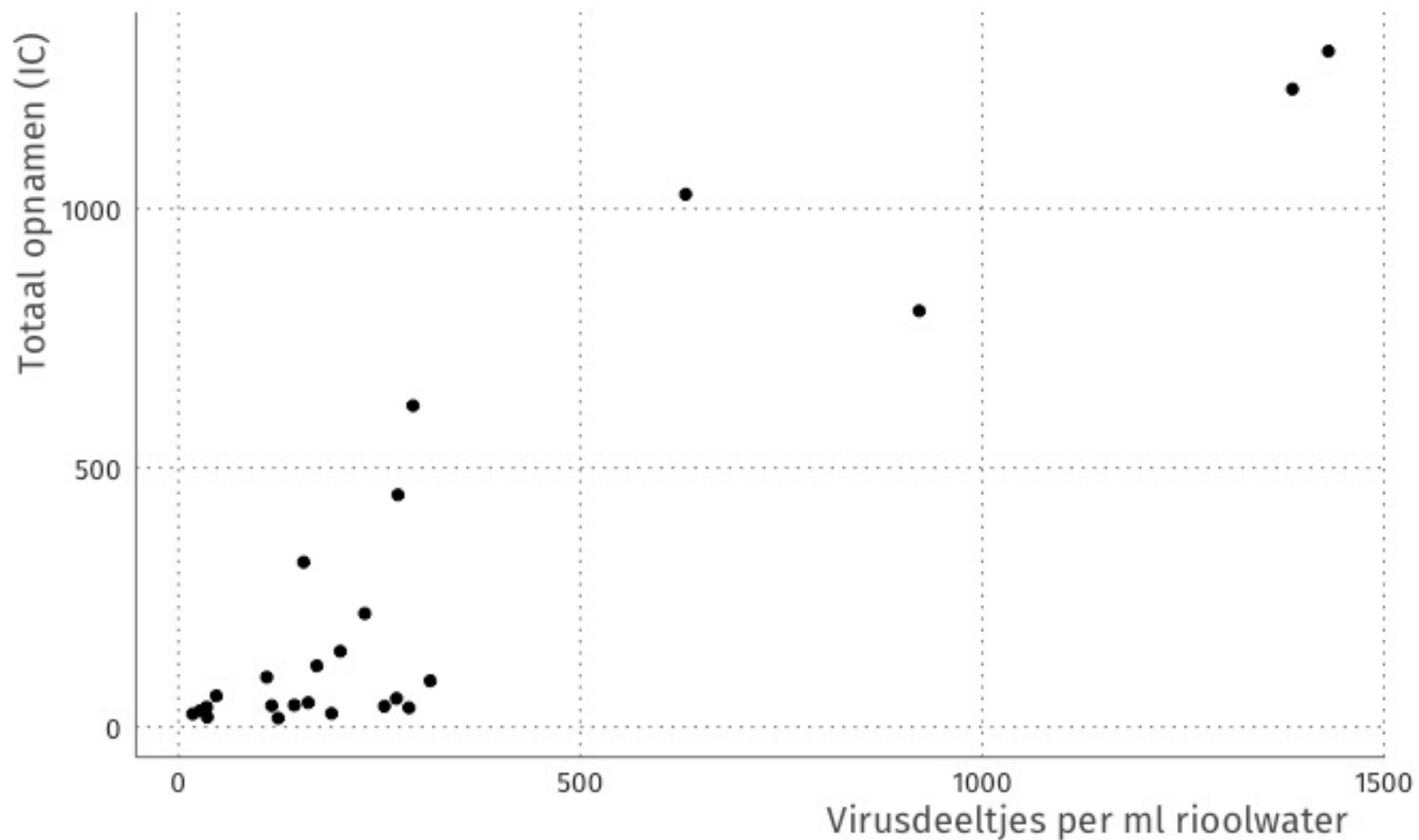
```
joined %>%
  filter(Type == "Totaal opnamen (IC)" |
         Type == "Virusdeeltjes per ml rioolwater") %>%
  ggplot(aes(Datum, Aantal, color = Type)) + geom_line() +
  scale_y_log10() +
  theme_fira() + scale_color_brewer(type = "qual")
```



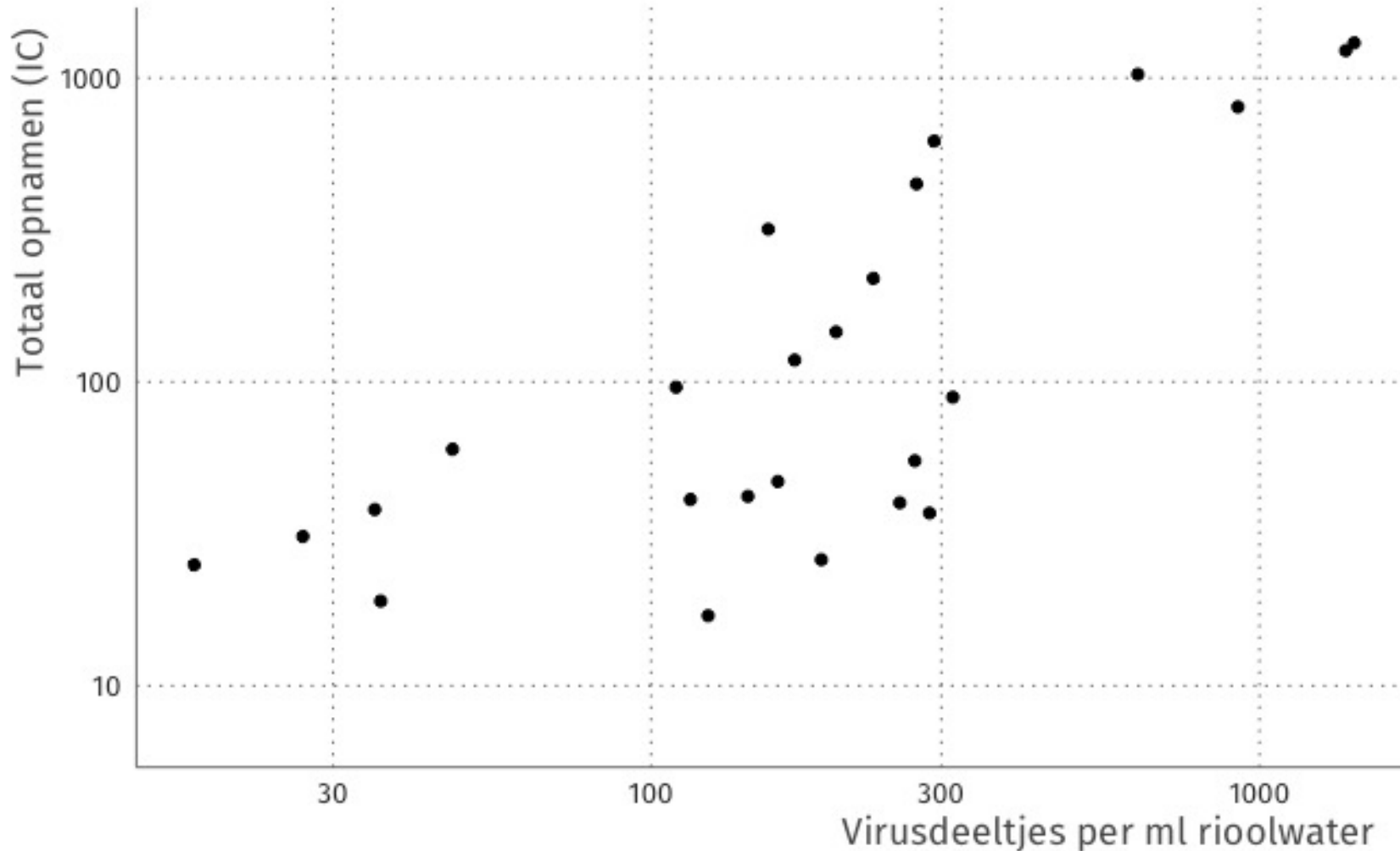
```
joined_wide <- joined %>%  
  filter(Type == "Totaal opnamen (IC)" |  
         Type == "Virusdeeltjes per ml rioolwater") %>%  
  pivot_wider(names_from = Type, values_from = Aantal)
```



```
joined_wide %>%  
ggplot(aes(`Virusdeeltjes per ml rioolwater`, `Totaal opnamen (IC)`)) +  
  geom_point() + theme_fira()
```

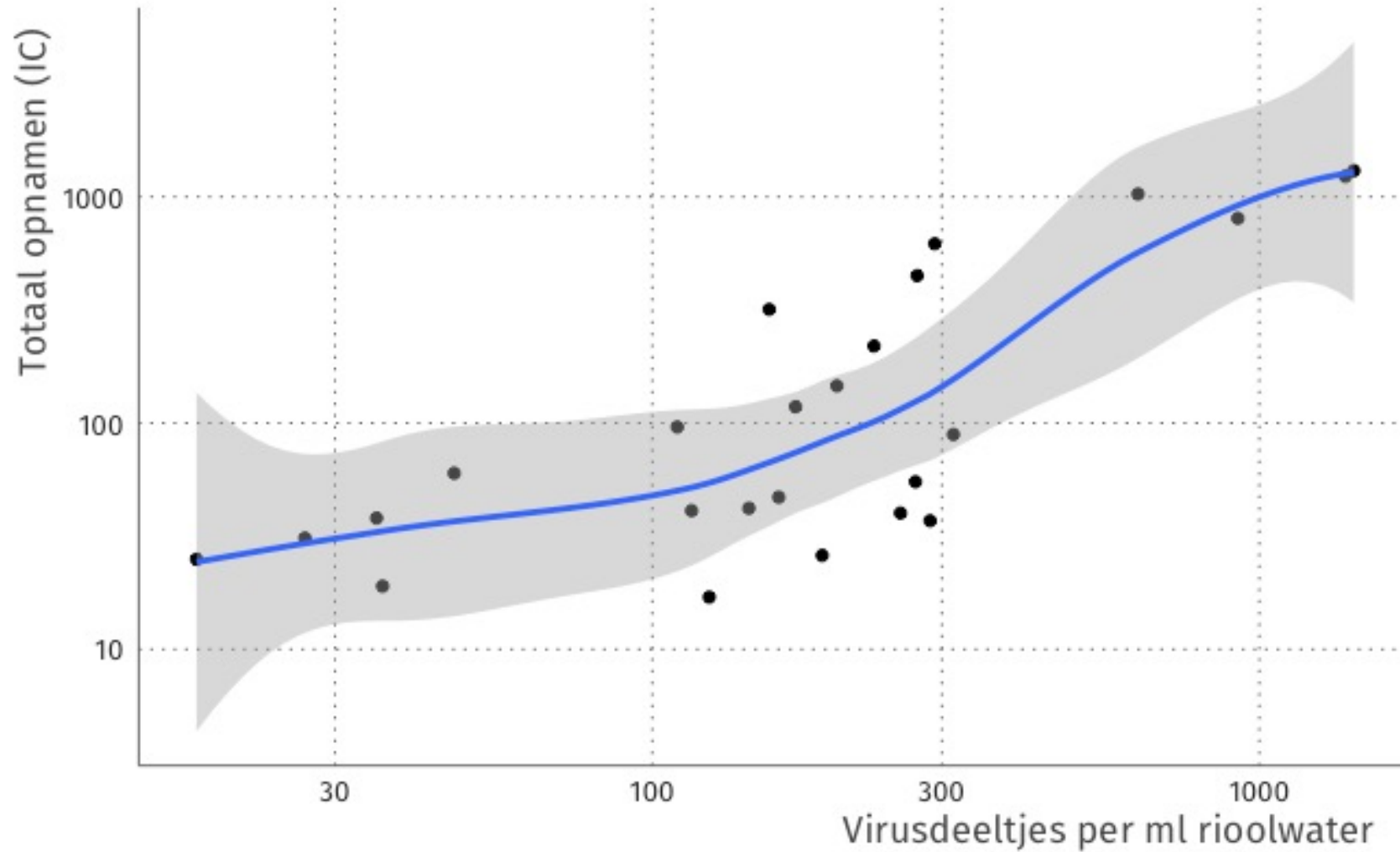


```
joined_wide %>%  
ggplot(aes(`Virusdeeltjes per ml rioolwater`, `Totaal opnamen (IC)`)) +  
  geom_point() + scale_y_log10() + scale_x_log10() + theme_fira()
```

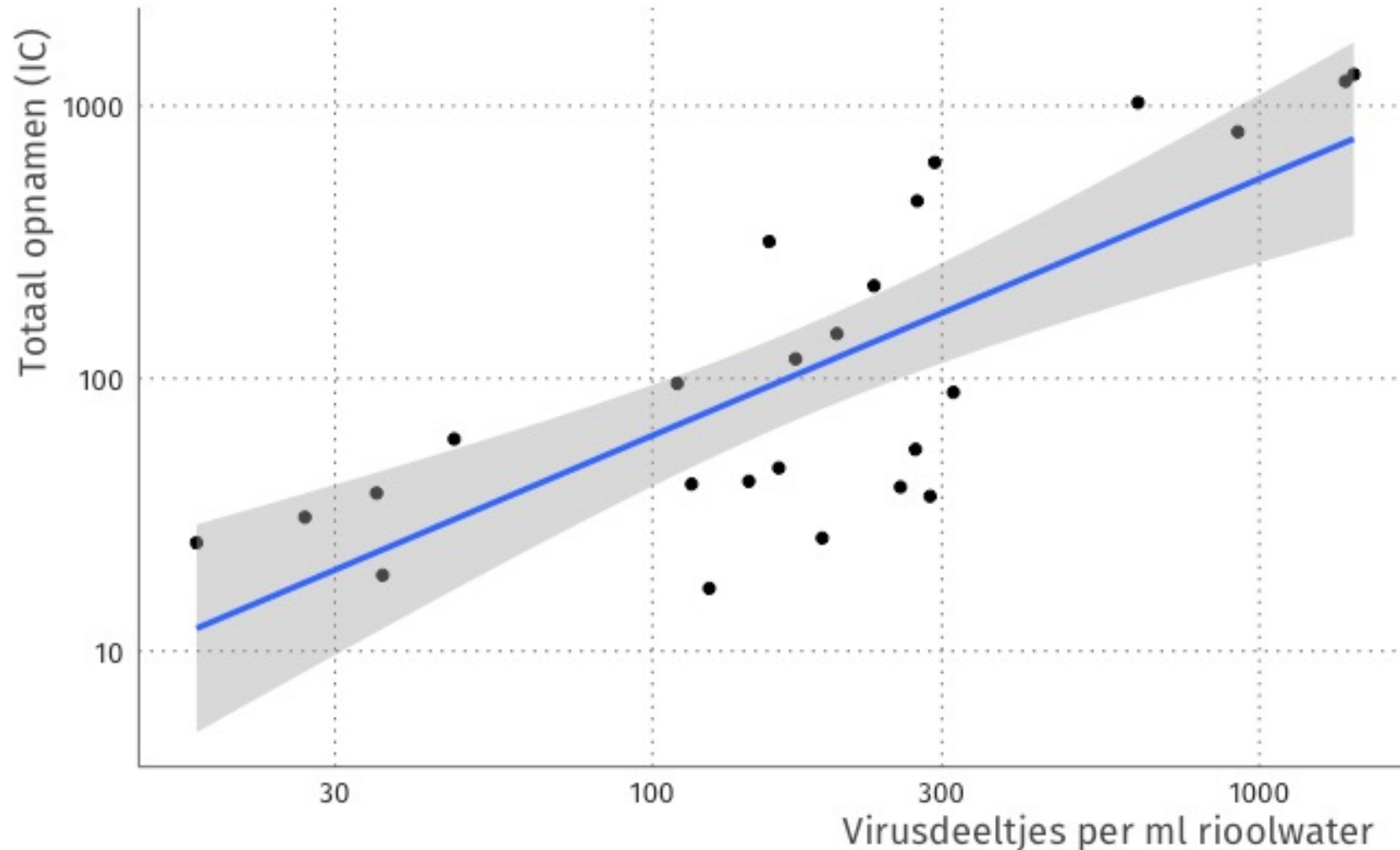


“straightening”

```
joined_wide %>%  
ggplot(aes(`Virusdeeltjes per ml rioolwater`, `Totaal opnamen (IC)`)) +  
  geom_point() + geom_smooth() +  
  scale_y_log10() + scale_x_log10() + theme_fira()
```




```
joined_wide %>%  
ggplot(aes(`Virusdeeltjes per ml rioolwater`, `Totaal opnamen (IC)`)) +  
  geom_point() + geom_smooth(method = "lm") +  
  scale_y_log10() + scale_x_log10() + theme_fira()
```



Looking at residuals

```
joined_wide <- joined_wide %>%  
  mutate(  
    log_icu = log(`Totaal opnamen (IC)` + 1),  
    log_sewage = log(`Virusdeeltjes per ml rioolwater` + 1))  
  
fit_lm <- lm(log_icu ~ log_sewage, data = joined_wide)  
  
joined_wide <- joined_wide %>%  
  mutate(log_icu_predicted = predict(fit_lm, newdata = .),  
         log_icu_residual = log_icu - log_icu_predicted)
```

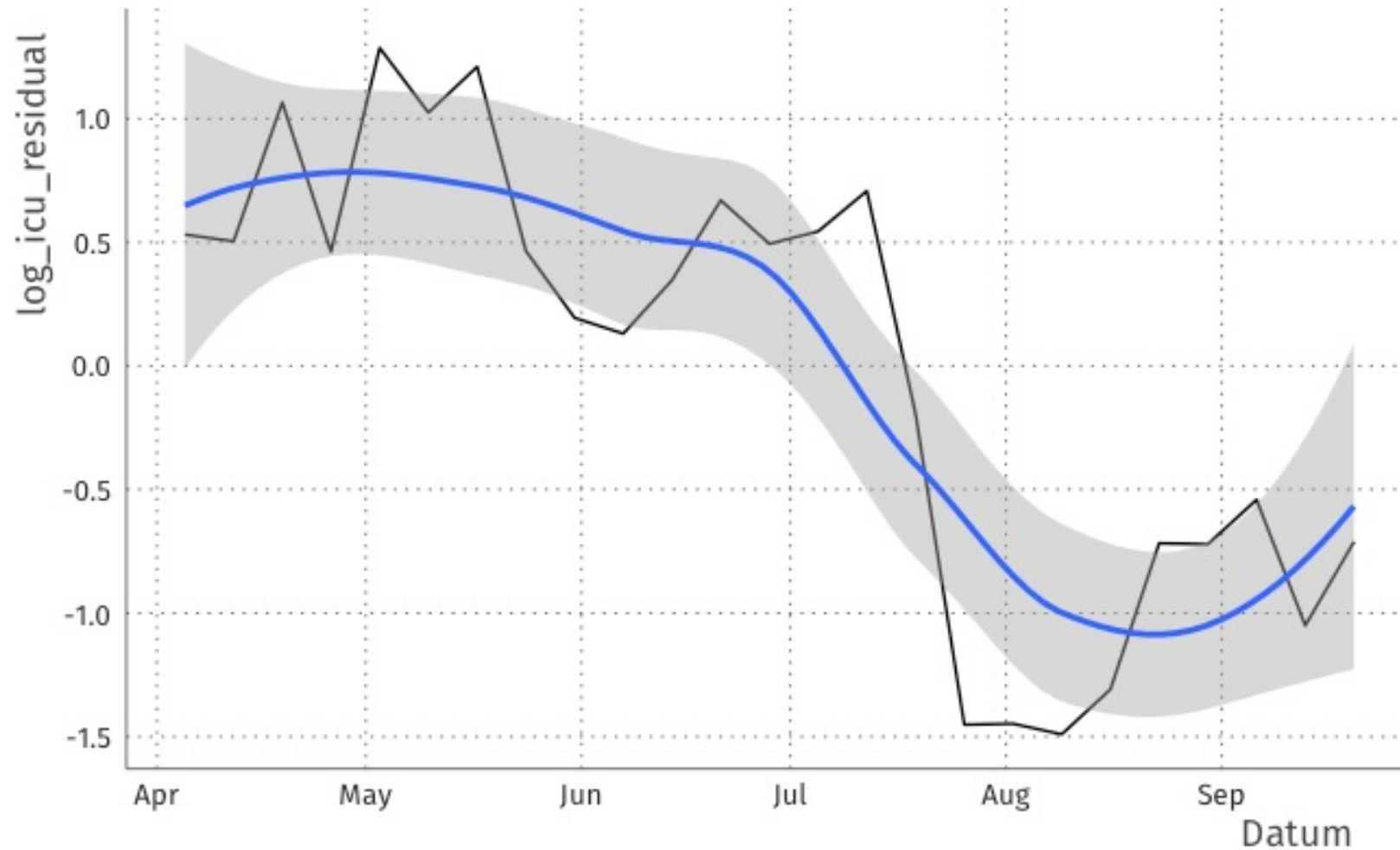


```
joined_wide %>%  
  filter(!is.na(log_icu_residual)) %>%  
  ggplot(aes(Datum, log_icu_residual)) +  
  geom_line() + theme_fira()
```



“flattening”

```
joined_wide %>%  
  filter(!is.na(log_icu_residual)) %>%  
  ggplot(aes(Datum, log_icu_residual)) +  
  geom_line() + geom_smooth() + theme_fira()
```



“flattening”

Using some standard plots for EDA

Visual EDA: two questions

- Variation: How are the features distributed?
 - Univariate
 - Specific: among people, among time, among flights
 - General: among the unit of measurement, among examples
 - In tabular data, generally one example/unit per row
- Association: What type of covariation occurs in the data?
 - Which features covary? When A is high, B is high?
 - Multidimensional, multivariate.



Visual EDA: two questions

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```
library(nycflights13)
```

```
flights
```

```
#> # A tibble: 336,776 x 19
```

```
#>   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
```

```
#>   <int> <int> <int>   <int>         <int>       <dbl>   <int>         <int>
```

```
#> 1  2013     1     1     517           515         2     830           819
```

```
#> 2  2013     1     1     533           529         4     850           830
```

```
#> 3  2013     1     1     542           540         2     923           850
```

```
#> 4  2013     1     1     544           545        -1    1004          1022
```

```
#> 5  2013     1     1     554           600        -6     812           837
```

```
#> 6  2013     1     1     554           558        -4     740           728
```

```
#> # ... with 3.368e+05 more rows, and 11 more variables: arr_delay <dbl>,
```

```
#> #   carrier <chr>, flight <int>, tailnum <chr>, origin <chr>, dest <chr>,
```

```
#> #   air_time <dbl>, distance <dbl>, hour <dbl>, minute <dbl>, time_hour <dtm>
```



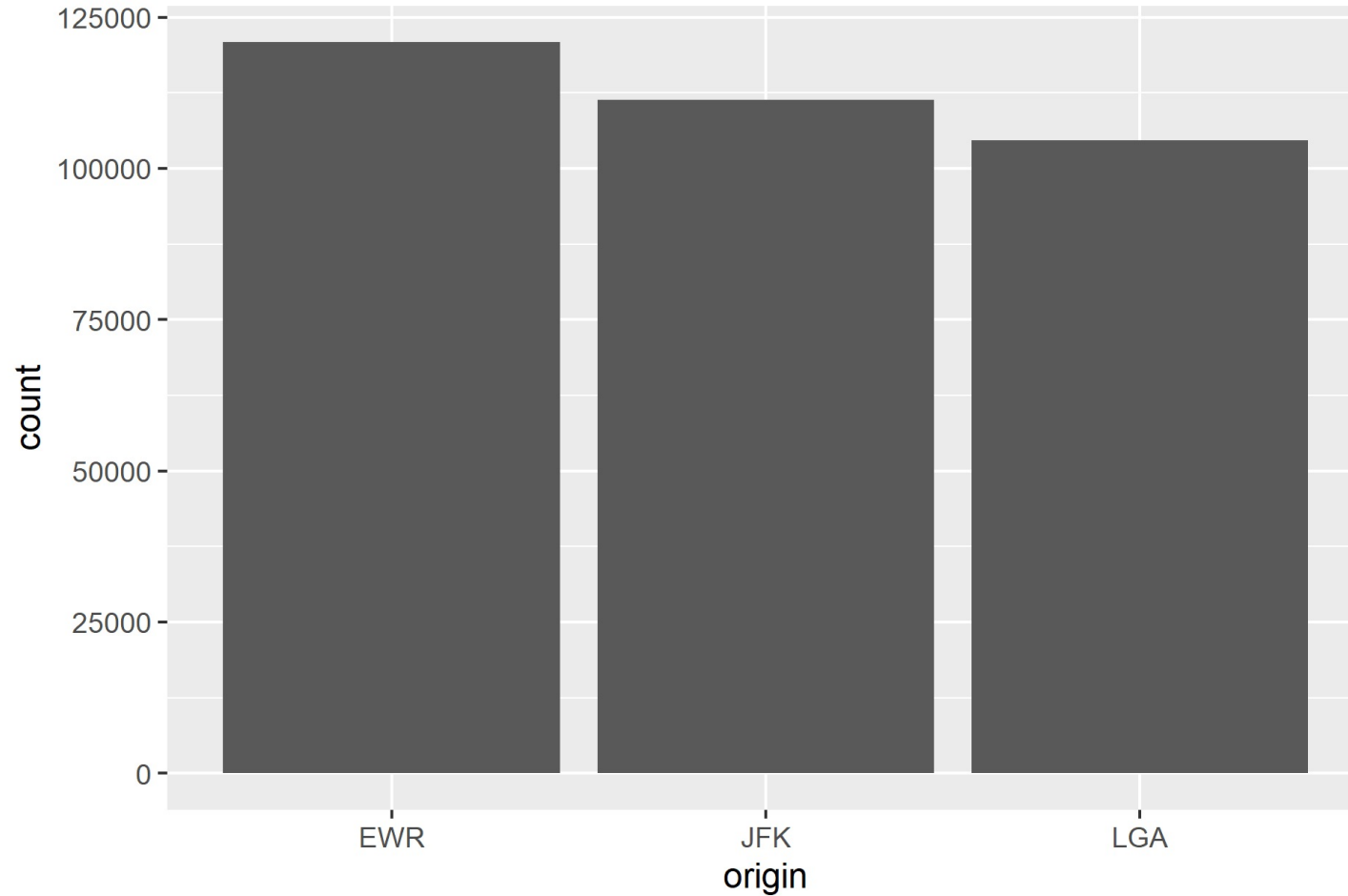
Bar chart

- X position: feature of interest
- Y position: count of occurrences (statistical transformation)
- Geom: bars/rectangle
- For categorical features e.g., flight origin



Bar chart

```
ggplot(data = flights) +  
  geom_bar(mapping = aes(x = origin))
```



Bar chart

- Fewest flights depart from LGA
- Almost 125000 flights from EWR



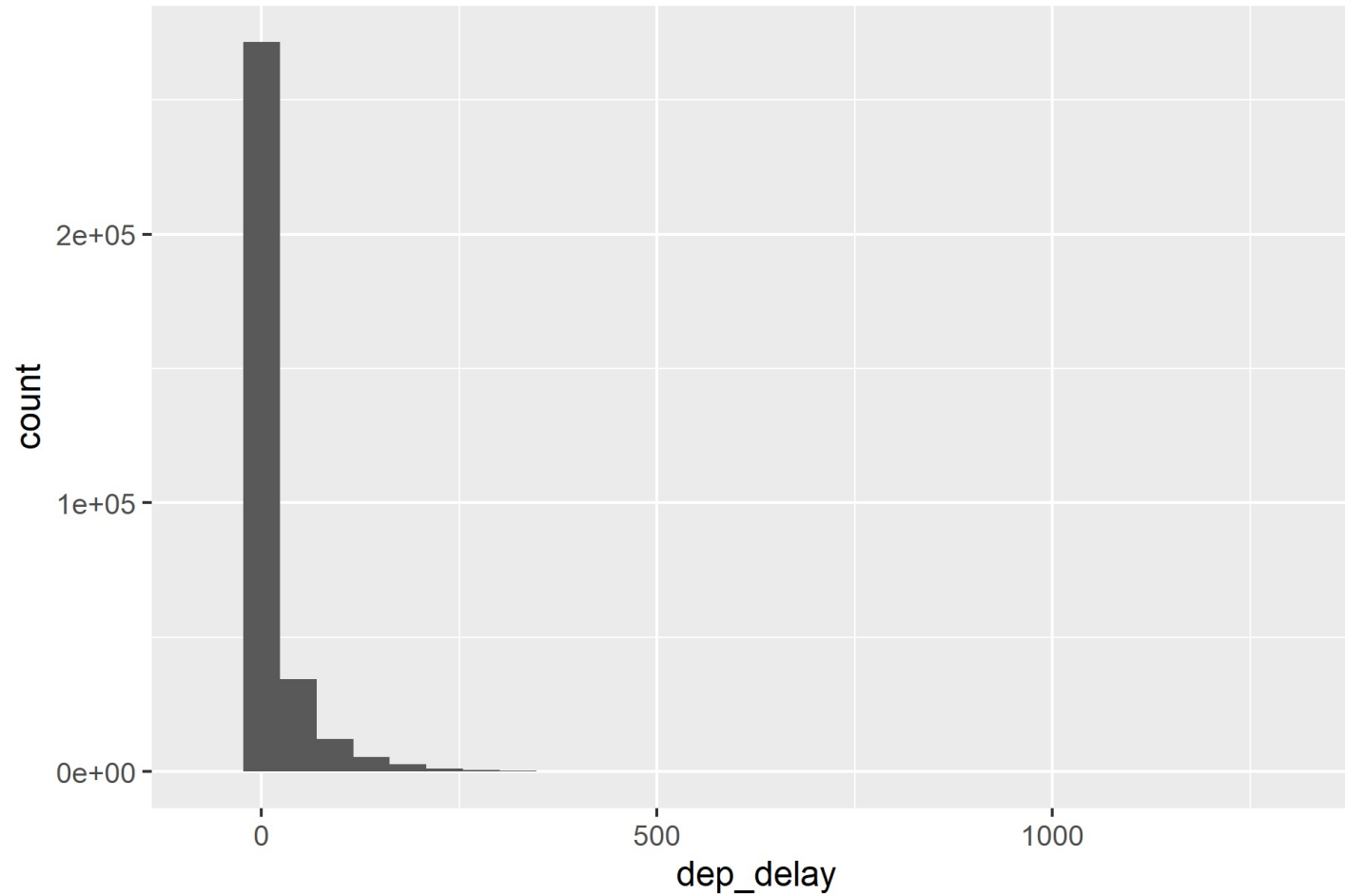
Histogram

- X position: feature of interest
 - Y position: count of occurrences **in bins** (statistical transformation)
 - Geom: bars/rectangle
-
- For continuous features, e.g., departure delay in minutes



Histogram

```
ggplot(data = flights) +  
  geom_histogram(mapping = aes(x = dep_delay))
```



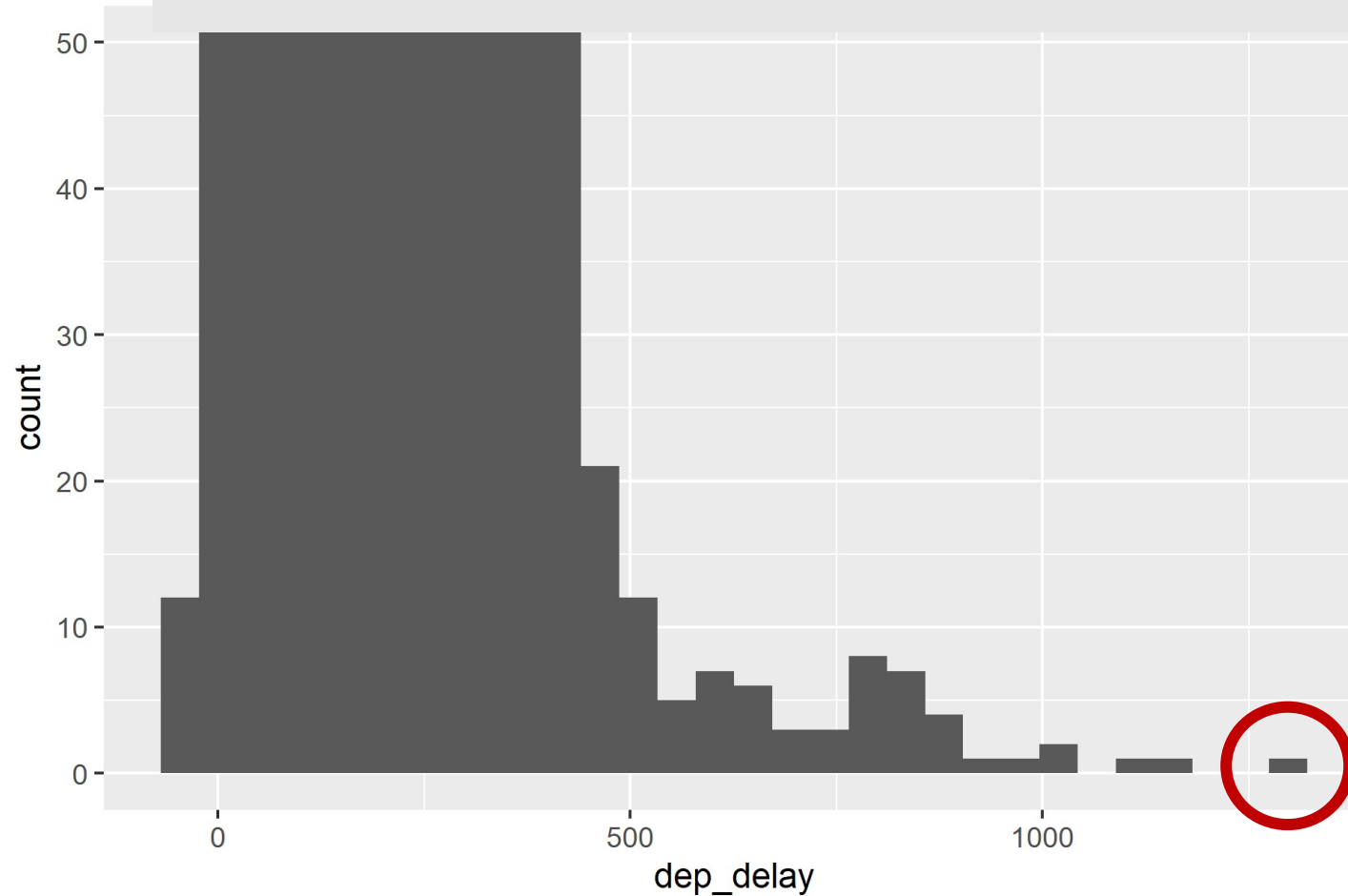
Histogram

- New question: why the strange distribution? What happens at the tail end?



Histogram (zoomed)

```
ggplot(data = flights) +  
  geom_histogram(mapping = aes(x = dep_delay)) +  
  coord_cartesian(ylim = c(0, 50))
```



That's a long delay!

Histogram

- Most flights are approximately on time
- There are even some flights that depart early
- There are a few flights with more than 1000 minutes delay



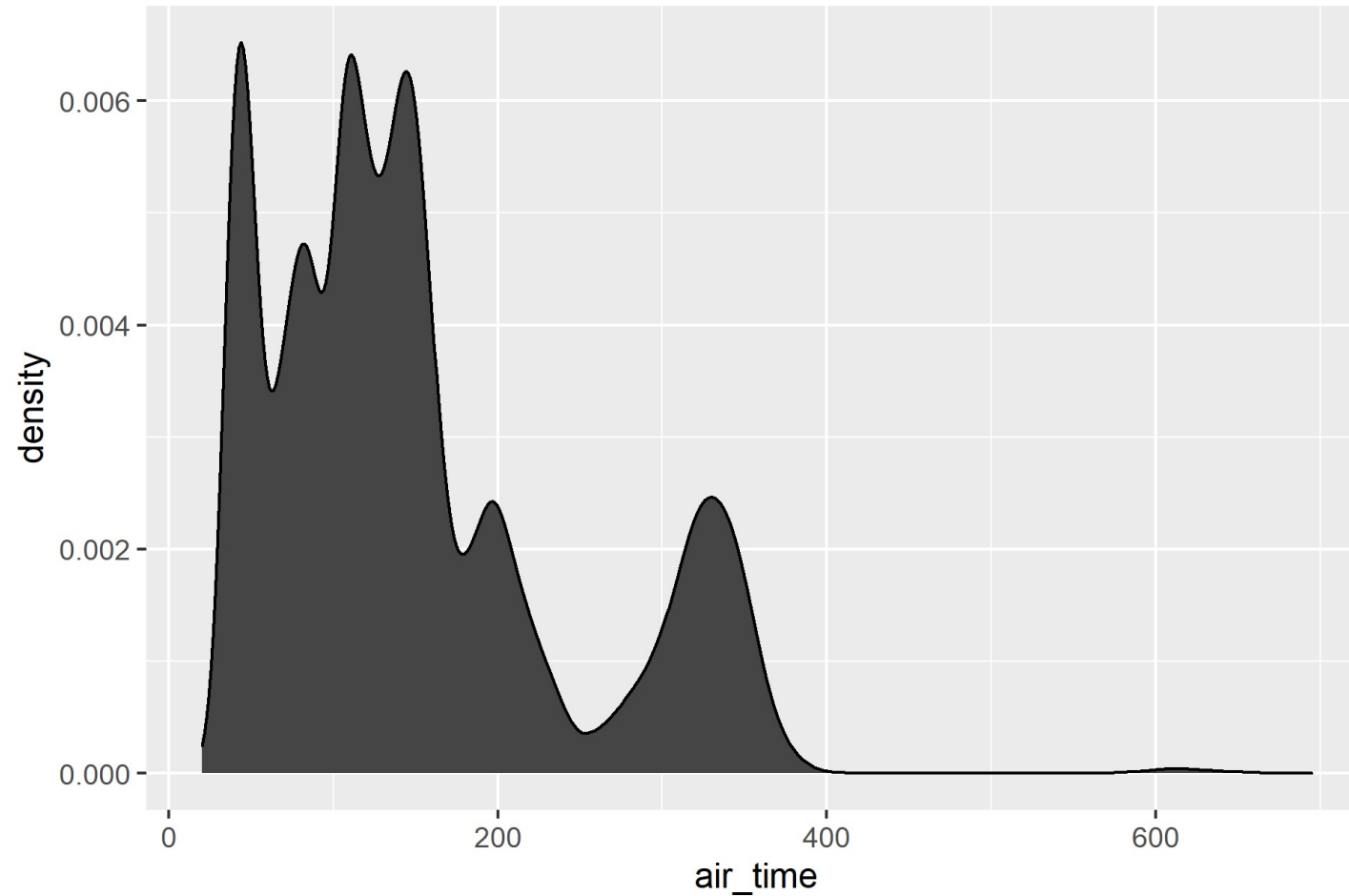
Density

- X position: feature of interest
 - Y position: density (statistical transformation, smoothed histogram)
 - Geom: polygon / line
-
- For continuous features, e.g., airtime in minutes



Density

```
ggplot(data = flights) +  
  geom_density(mapping = aes(x = air_time))
```



Density

- Most flights from NY are under 200 minutes (3.3 hours)
 - Few flights are between 200 and 250 minutes
 - Quite some flights are between 250 and 400 minutes
 - Some flights are over 600 minutes (weird bump? Remember!)
-
- Air times are not normally distributed!



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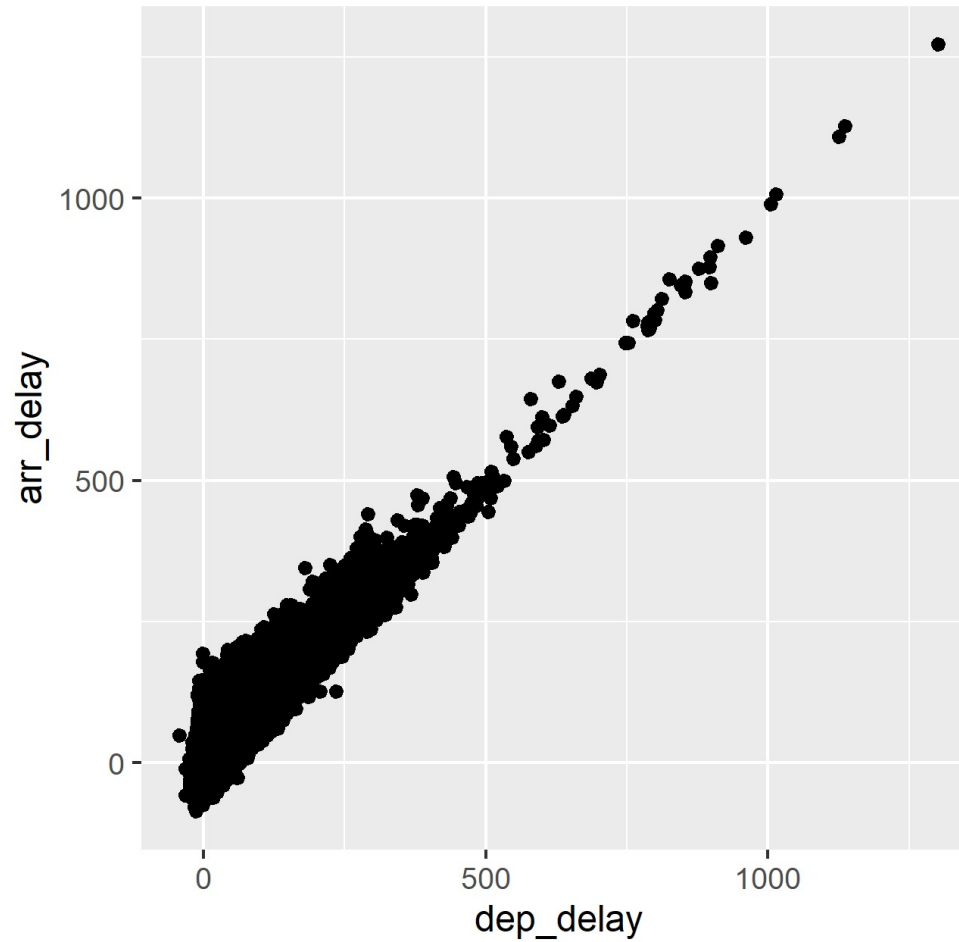
Scatter plot

- X position: feature A of interest (in regression, generally predictor)
 - Y position: feature B of interest (in regression, outcome)
 - Geom: points/dots
-
- For continuous features, e.g., departure delay in minutes and arrival delay in minutes



Scatter plot

```
ggplot(data = flights) +  
  geom_point(aes(x = dep_delay, y = arr_delay))
```



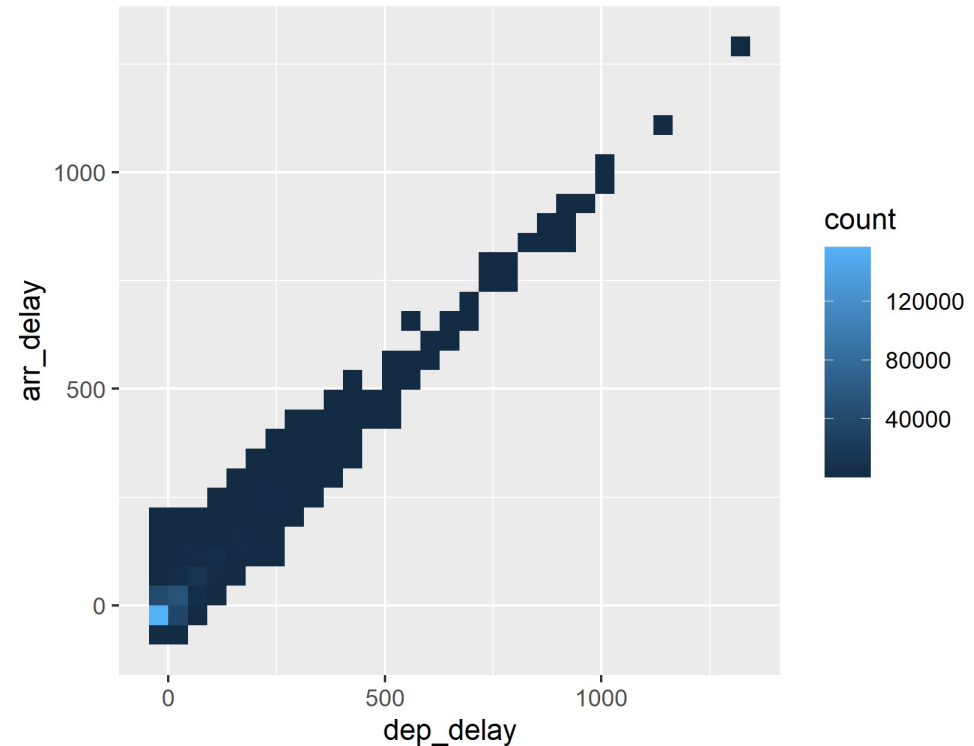
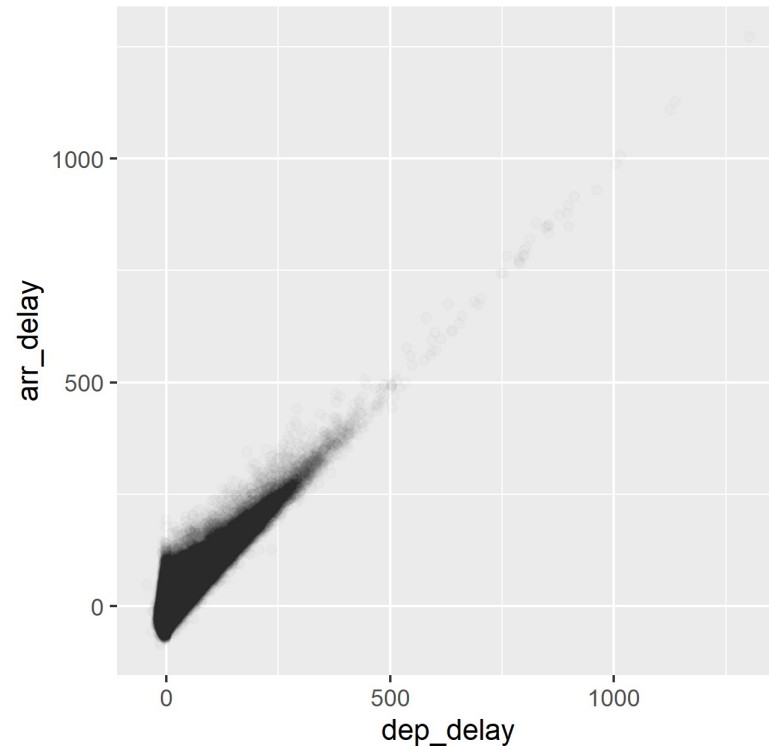
Scatter plot

- Departure and arrival delay correlate strongly
- Arrival delay is generally higher than departure delay
- Only a few delays are above 500 minutes

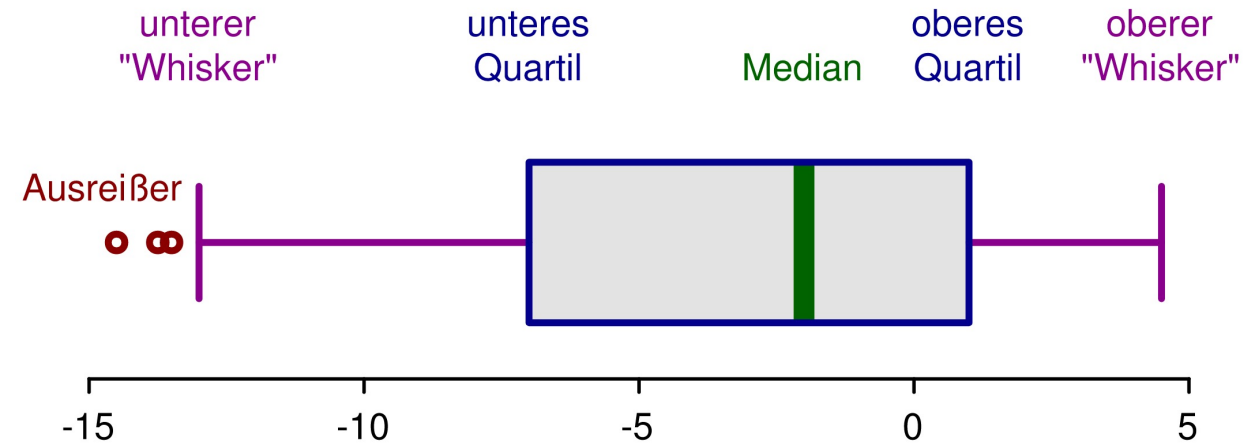


Solutions to overplotting

- Transparency (alpha) or binning (geom_bin2d)



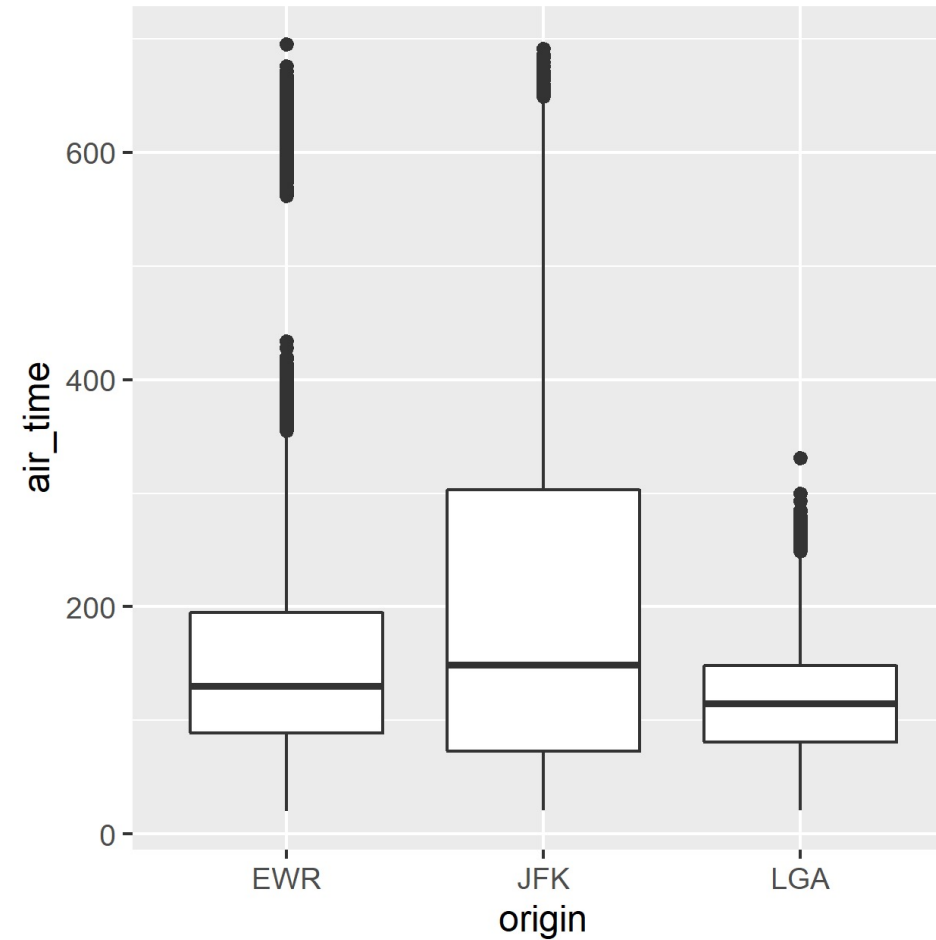
Box plot (Tukey)



- X position: feature A of interest
- Y position: feature B of interest
- Geom: rectangles for box, lines (whiskers), points for outliers
- Statistical transformations:
 - median, 25% and 75% percentile (inter quartile range, IQR), $1.5 \times \text{IQR}$ for “whiskers”
- Continuous vs. categorical, e.g., origin and airline

Box plot

```
ggplot(data = flights) +  
  geom_boxplot(aes(x = origin, y = air_time))
```

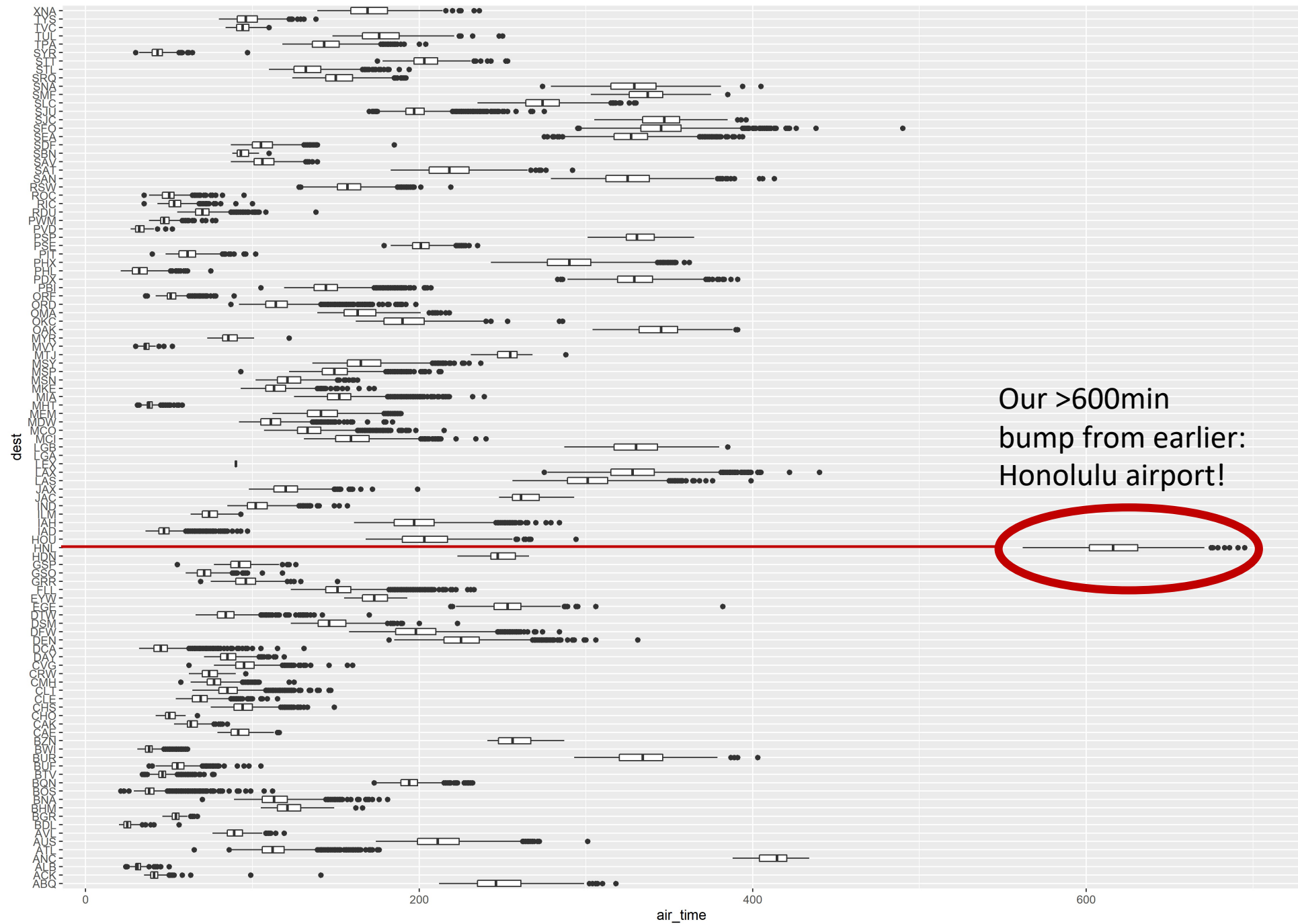


Box plot

- Flights from LGA do not have the >600 minute bump
 - Flights from JFK take longest, on average (median)
 - Flight times from JFK have largest IQR
-
- Let's look at destination rather than origin?

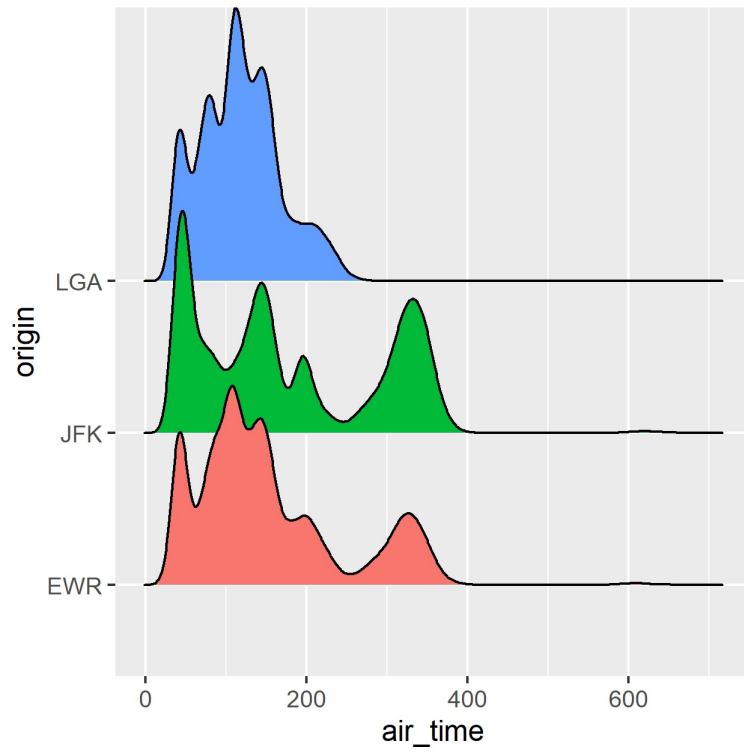


Box plot by destination



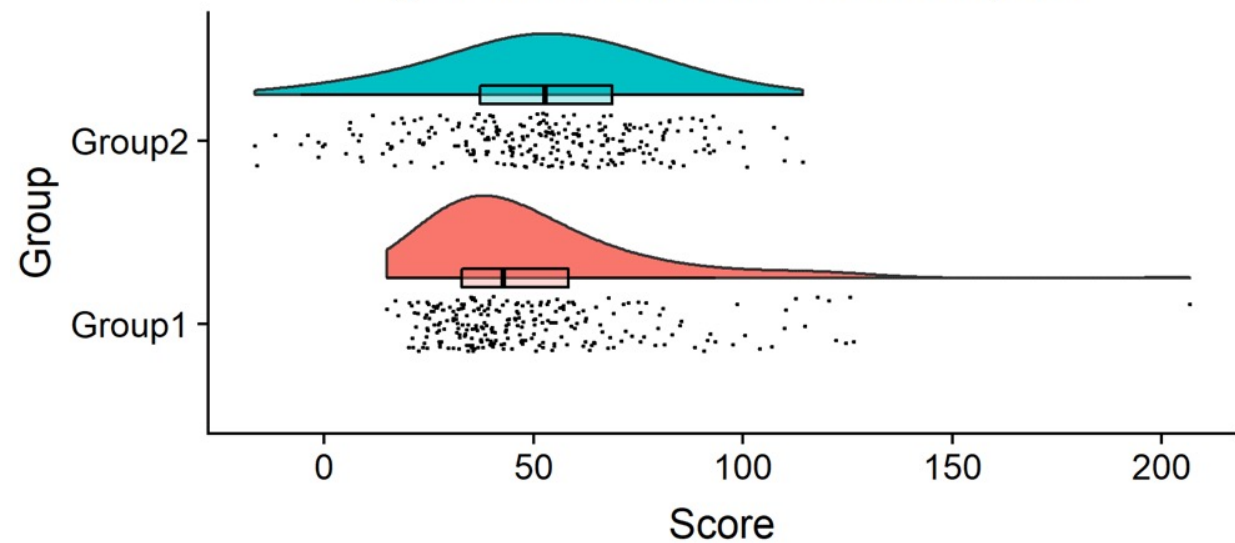
Many options available for categorical vs continuous

ggridges (ridgeline plot)



Raincloud plot

Figure R5: Raincloud Plot w/Boxplots



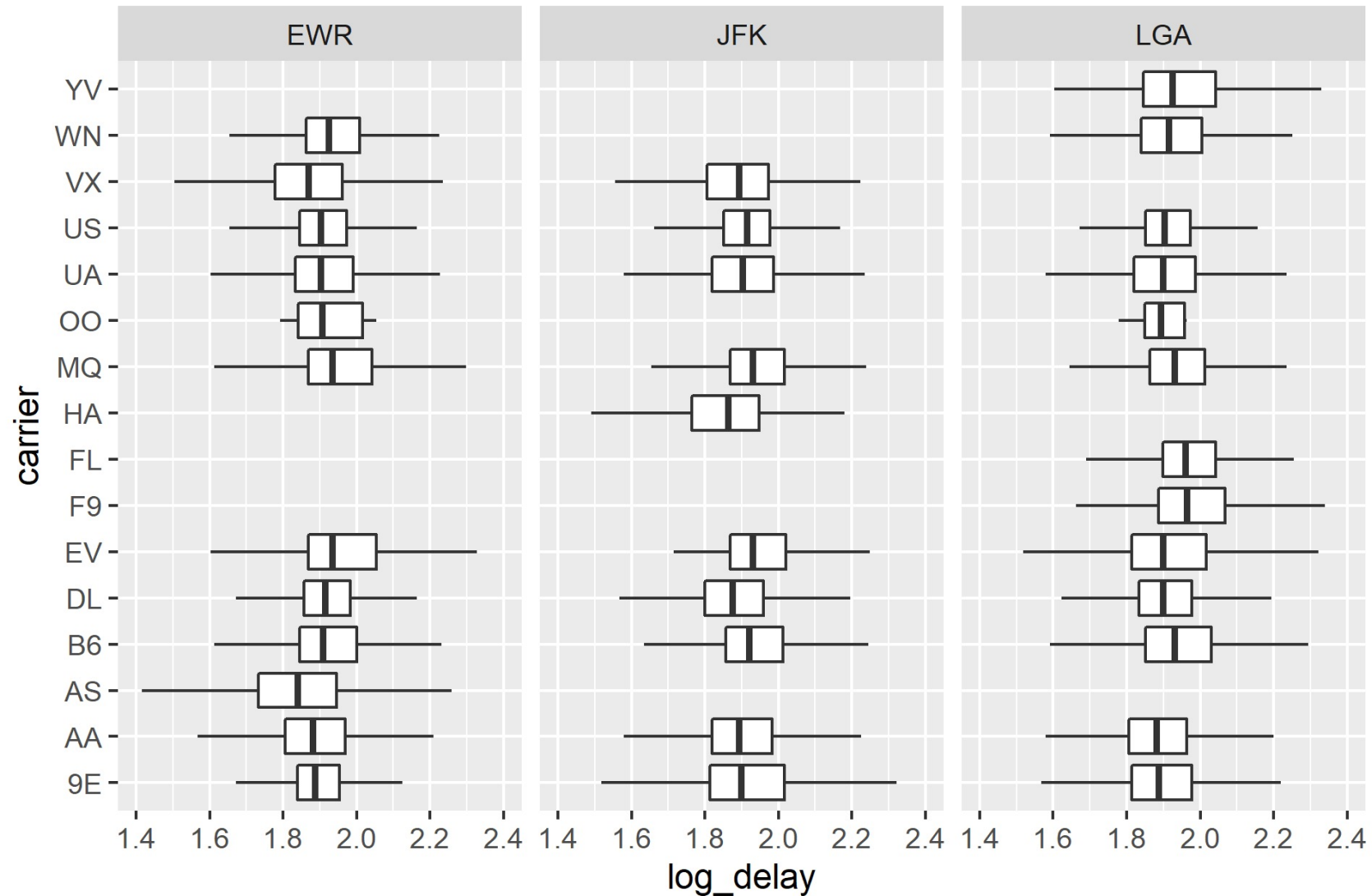
Facets

- Faceting a plot means creating a subplot per category
- Also called “small multiples” (tufte)
- Allows for another categorical variable in the visualisation
- Danger: clutter

+ `facet_wrap(~feature)`



Boxplot of (log) delay by carrier, faceted by origin



Facets

- Some carriers do not fly from some airports (missing data)
- Delays are quite similar across origin airports and carriers

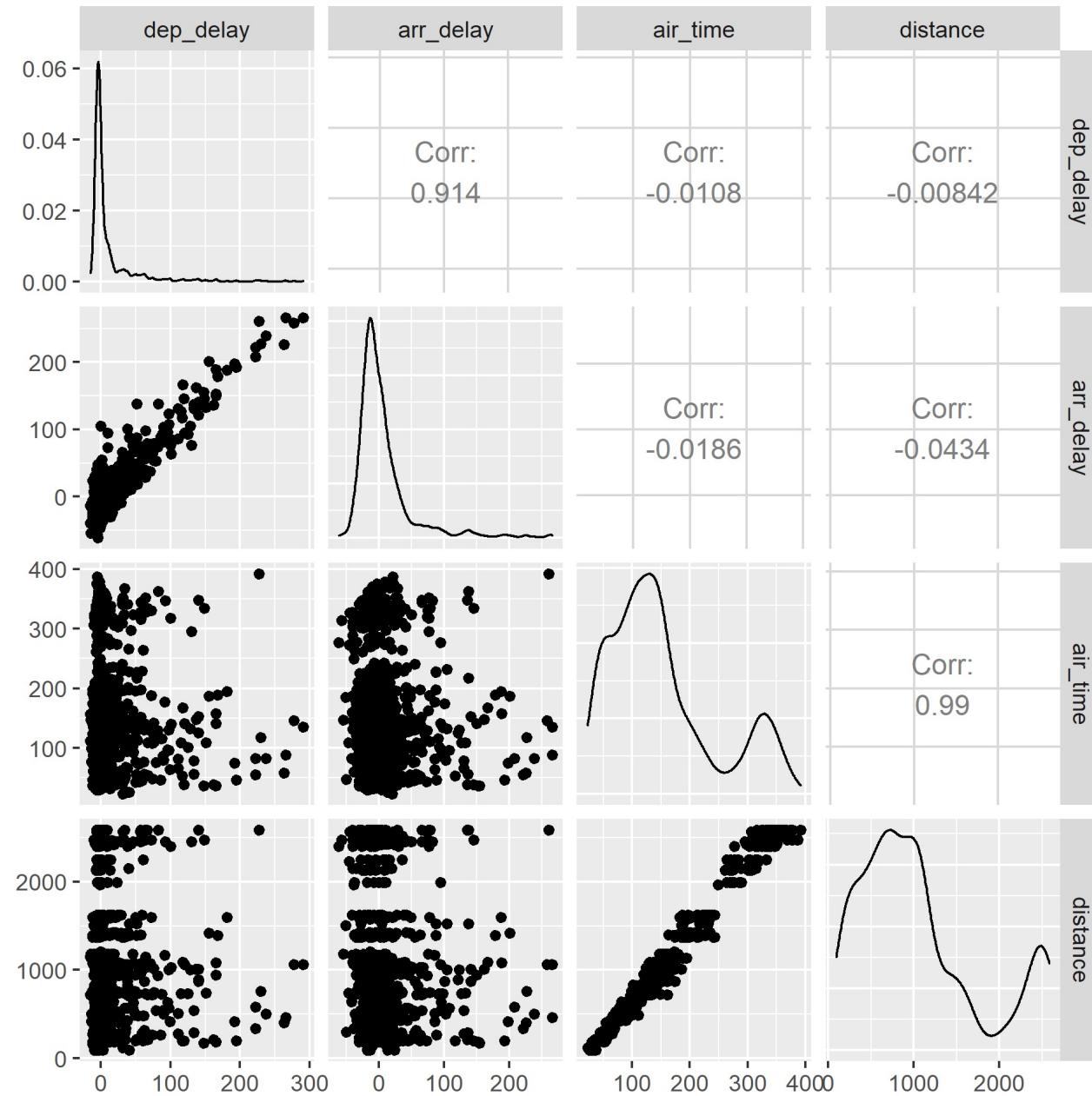


Pairs plot

- Pairs plot: a scatter plot for each variable
- Multivariate
- Difficult in ggplot: facet by variable (not a feature in the original dataset)
- Data frame needs to be
- Package available: GGally (ggpairs)



Pairs plot



Pairs plot

- Air time and distance correlate strongly
 - Departure and arrival delay correlate strongly
 - Air time / distance do not correlate with delay
-
- In other words: distance cannot predict delays well



Conclusion

- Exploration is key to understanding things you did not know already
- EDA walks a fine line between seeing useful and less useful things (overfitting, if you like)
- Some useful principles are:
 - Peng's checklist
 - Understanding what you're seeing
 - Finding interesting comparisons
 - "Straightening and flattening" (using models and residuals)
 - Standard graphs to look at variation and association

